



Evaluation of 14 nonlinear deformation algorithms applied to human brain MRI registration

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Abstract

All fields of neuroscience that employ brain imaging need to communicate their results with reference to anatomical regions. In particular, comparative morphometry and group analysis of functional and physiological data require coregistration of brains to establish correspondences across brain structures. It is well established that linear registration of one brain to another is inadequate for aligning brain structures, so numerous algorithms have emerged to nonlinearly register brains to one another. This study is the largest evaluation of nonlinear deformation algorithms applied to brain image registration ever conducted. Fourteen algorithms from laboratories around the world are evaluated using 8 different error measures. More than 45,000 registrations between 80 manually labeled brains were performed by algorithms including: AIR, ANIMAL, ART, Diffeomorphic Demons, FNIRT, IRTK, JRD-fluid, ROMEO, SICL, SyN, and four different SPM5 algorithms (“SPM2-type” and regular Normalization, Unified Segmentation, and the DARTEL Toolbox). All of these registrations were preceded by linear registration between the same image pairs using FLIRT. One of the most significant findings of this study is that the relative performances of the registration methods under comparison appear to be little affected by the choice of subject population, labeling protocol, and type of overlap measure. This is important because it suggests that the findings are generalizable to new subject populations that are labeled or evaluated using different labeling protocols. Furthermore, we ranked the 14 methods according to three completely independent analyses (permutation tests, one-way ANOVA tests, and indifference-zone ranking)

and derived three almost identical top rankings of the methods. ART, SyN, IRTK, and SPM's DARTEL Toolbox gave the best results according to overlap and distance measures, with ART and SyN delivering the most consistently high accuracy across subjects and label sets. Updates will be published on the <http://www.mindboggle.info/papers/> website.

Introduction

Brain mapping – mapping the structures, physiology, functions, and connectivity of brains in individuals and in different populations – is possible due to a diverse but often disconnected array of brain imaging technologies and analysis methods. To make the best use of brain image data, researchers have attempted for over 40 years to establish a common reference frame such as a three-dimensional coordinate or labeling system to consistently and accurately communicate the spatial relationships within the data (Talairach and Szikla, 1967, Talairach and Tournoux, 1988, Drury et al., 1996, Fischl et al., 1999, Clouchoux et al., 2005). A common reference frame helps us to:

1. communicate and compare data (across subjects, time, conditions, and image types),
2. classify data (by meaningful spatial positions or extent), and
3. find patterns in data (to infer structural or functional relationships).

These three benefits are contingent on one serious premise: positions and sizes in one brain must correspond to positions and sizes in another brain to make comparisons.

This premise almost universally does not hold when brain image data are compared across individuals. The noise that this introduces is often accepted by researchers who generally assume that if they have found corresponding features across two brains, the intervening points between those features correspond to one another as well. Brains are so variable in shape that there simply may not exist a point-to-point correspondence across any two brains, or even in the same brain over time.

Explicit manual labeling of brain regions is the preferred approach for establishing anatomical correspondence, but it is too prohibitive in terms of time and resources, particularly in cases where neuroanatomists are not available, in intraoperative or other time-sensitive scenarios, and in high-throughput environments that need to process dozens to thousands of brain images.¹

Automatically determining anatomical correspondence is almost universally done by registering brains to one another or to a template. There has been a proliferation of different approaches to perform image registration that demands a comparison to guide choices regarding algorithms, software implementation, setup and parameters, and data preprocessing options. To better enable individuals to make these choices, the Valmet software tool (<http://www.ia.unc.edu/public/valmet/>) (Gerig et al., 2001) and the Non-rigid Image Registration Evaluation Project (NIREP) (<http://www.nirep.org>) were developed. The Windows-based Valmet was in 2001 the first publicly available software tool for measuring (as well as visualizing) the differences between corresponding image segmentations, but has received only one minor update since 2001 (in 2004). It uses several algorithms to compare segmentations: overlap ratio, Hausdorff distance, surface

distance, and probabilistic overlap. The NIREP project “has been started to develop, establish, maintain, and endorse a standardized set of relevant benchmarks and metrics for performance evaluation of nonrigid image registration algorithms.” The initial phase of the project will include 16 manually labeled brain images (32 labeled regions in 8 men and 8 women) and four evaluation metrics: 1. relative overlap (equivalent to the “union overlap” defined in the Materials and methods section), 2. variance of the registered intensity images for an image population, 3. inverse consistency error between a forward and reverse transformation between two images, and 4. transitivity (how well all the pairwise registrations of the image population satisfy the transitivity property).

In this study we set out to evaluate what we believe are the most important nonlinear deformation algorithms that have been implemented in fully automated software programs and applied to human brain image registration. We measure accuracy at the scale of gross morphological structures (gyri, sulci, and subcortical regions) acquired by magnetic resonance imaging (MRI). There have been two significant prior studies that compared more than three nonlinear deformation algorithms for evaluating whole-brain registration.

The first was communicated in a series of publications by Hellier et al., 2001a, Hellier et al., 2002, Hellier et al., 2003); they compared five different fully automated nonlinear brain image registration software programs using the same set of quantitative measures. These included global measures comparing 17 deformed MRI source images and one target image: average brain volume, gray matter overlap, white matter overlap, and correlation of a measure of curvature, and local measures of distance and shape between corresponding principal sulci. Our study includes a version of each of the five methods and is different primarily because (1) all tests were conducted by a single individual (the first author) who had not authored any of the software packages, but received guidance from the principal architects of the respective algorithms, (2) its focus is on manually labeled anatomical regions, and (3) each and every brain was used as a source and as a target for registration rather than selecting a single target.

The second is a recent paper (Yassa and Stark, 2009) that compares nonlinear registration methods applied to regions in the medial temporal lobe; six of the methods are fully automated and two are semi-automated (requiring manual identification of landmarks). They apply these methods either to manually labeled brain regions, to weighted masks for these regions, or to the original unlabeled brains, as in our study. The four methods that they applied to unlabeled brains (and evaluated on regions in the medial temporal lobe) are the Talairach piecewise linear approach and three SPM programs (included in our study). Registering labeled regions obviously requires that the regions be labeled; their ROI-AL approach ‘labels to register’ rather than ‘registers to label’ or ‘registers without labels.’ They used two evaluation measures on pairs of images (20 MRI volumes total): an overlap measure (equivalent to the “target overlap” defined in the Materials and methods section) and a measure of blur in a group average of coregistered images.

What sets our study apart from both of these prior studies is the unparalleled scale and thoroughness of the endeavor:

- over 14 nonlinear algorithms
- each algorithm applied at least 2,168 times (over 45,000 registrations total)

- 80 manually labeled brain images
- 4 different whole-brain labeling protocols (56 to 128 labeled regions)
- 8 different evaluation measures
- 3 independent analysis methods.

This study evaluates 15 registration algorithms, one linear (FLIRT) and 14 nonlinear: AIR, ANIMAL, ART, Diffeomorphic Demons, FNIRT, IRTK, JRD-fluid, ROMEO, SICLE, SyN, and four different SPM5 algorithms (“SPM2-type” and regular Normalization, Unified Segmentation, and the DARTEL Toolbox; DARTEL was also run in a pairwise manner and all four SPM algorithms were run with and without removal of skulls from the images). The linear algorithm was included as an initialization step to establish a baseline prior to applying the nonlinear algorithms. Comparisons among the algorithms and their requirements are presented in Table 1 and in the Appendix B, software commands are in Supplementary section 7, and brief descriptions are in Supplementary section 8. Many of them are in common use for registering structural MRIs to each other or to templates for neuromorphometric research or as an intermediary to compare functional or physiological data (Gholipour et al., 2007), but some of them exist only as pre-release code made available by their respective authors for this study. See the “Algorithms excluded from the study” section in the Discussion for algorithms excluded from the study. Additional materials and updated information will be made publicly available via the website <http://www.mindboggle.info/papers/> ⁷.

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Section snippets

Materials and methods

In this section, we first briefly describe the acquisition and preparation of the brain image and label data. Then we outline the preprocessing (brain extraction and formatting), linear registration, and nonlinear registration stages applied to the data, our evaluation measures, and our analysis methods. The first author performed these latter steps on an OSX system (Mac Pro 2-Quad-Core (8-processor) Intel Xeon, 3GHz, 6GB RAM) with a 10.4 operating system, except where noted (see ...

Results

Results for the initial run are in Supplementary section 1, for the trivial case, where each brain was registered to itself, are in Supplementary section 2, volume similarity results are in Supplementary

section 4, and distance error results are in Supplementary section 5. ...

Discussion

This study evaluates 15 registration algorithms (one linear, 14 nonlinear) based primarily on overlap measures of manually labeled anatomical regions. The scale and thoroughness are unprecedented (over 45,000 registrations, 80 manually labeled brain images representing 4 different labeling protocols, 8 different evaluation measures, and 3 independent analysis methods). We hope that the method of evaluation as well as the results will be useful to the neuroscience community. As they become ...

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